



Quartiles, Deciles, and Percentiles

There are measures of location that divide a set of observations into equal parts. These measures include ***quartiles, deciles, and percentiles***.

Quartiles divide a set of observations into four equal parts. We call the middle value of a set of data arranged from smallest to largest the median. That is, 50 percent of the observations are larger than the median and 50 percent are smaller. The median is a measure of location because it pinpoints the center of the data.

In a similar fashion quartile divide a set of observations into four equal parts. The first quartile, usually labeled Q_1 , is the value below which 25 percent of the observations occur and the third quartile, usually labeled Q_3 , is the value below which 75 percent of the observations occur. Q_2 is the median. The values corresponding to Q_1 , Q_2 , and Q_3 divide a set of data into four equal parts.

In a similar fashion decile divide a set of observations into 10 equal parts and percentiles into 100 equal parts.

► Median like measures-

❖ Quartiles

❖ Deciles

❖ Percentiles

The price of shares for a sample of five companies are:

21, 25, 19, 20, 22

Arranging the data in ascending order gives:

19, 20, 21, 22, 25

The heights of four basketball players, in inches are:

76, 73, 80, 75

Arranging the data in ascending order gives:

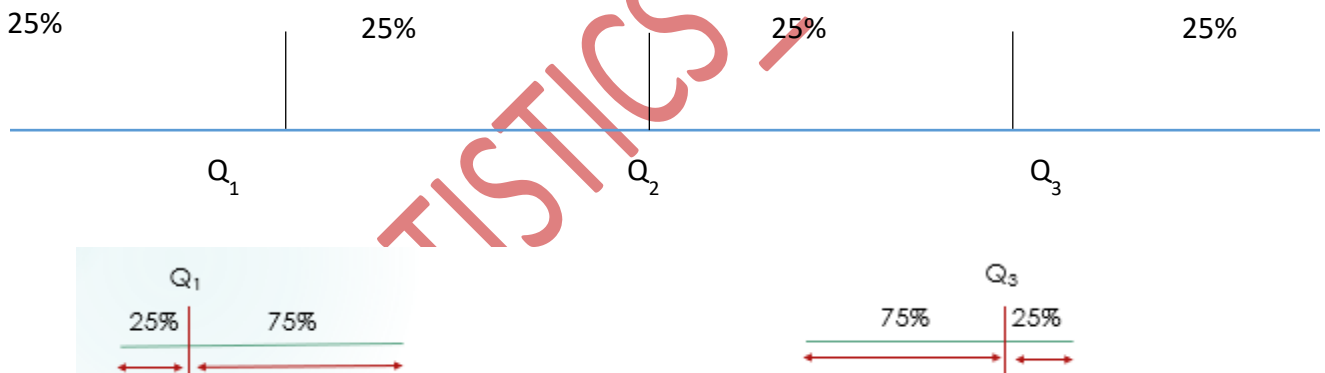
73, 75, 76, 80

Thus, the median is 75.5

Median like measures:

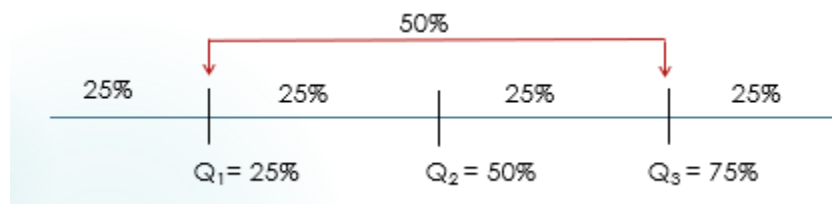
Quartiles:

Quartiles Divides the whole distribution into 4 equal parts



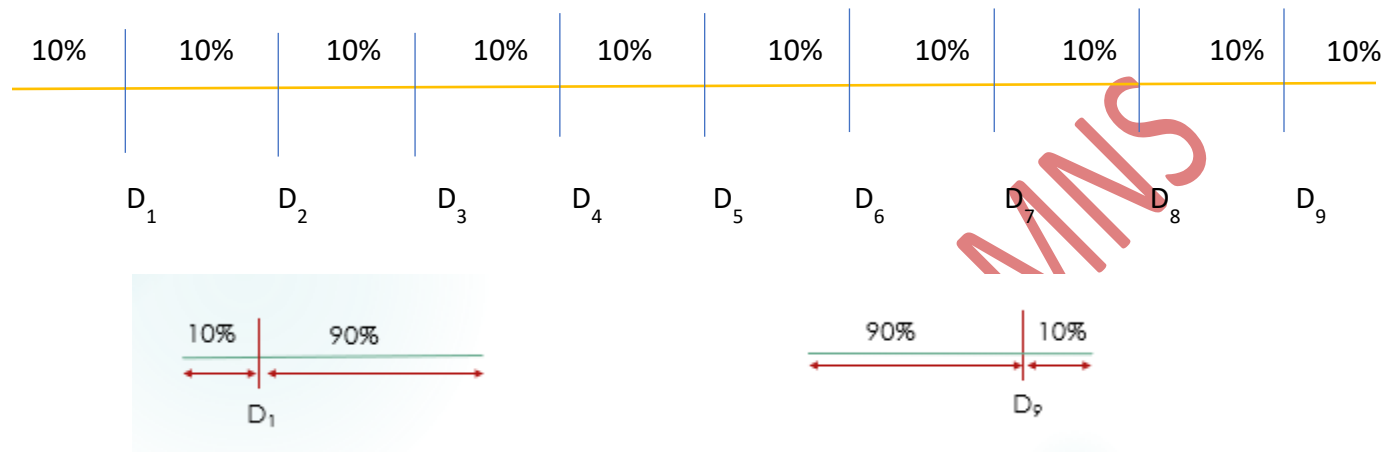
$$\text{Interquartile Range (IQR)} = Q_3 - Q_1$$

Interquartile Range is the middle 50%, which trims off the 25% on both sides. Therefore, it is not affected by the outliers like we see in case of normal Range.



Deciles:

Deciles divide the whole distribution into 10 equal parts



Percentiles:

Percentiles divide the whole distribution into 100 equal parts. A percentile, p , is a measure used to indicate the value below which a given percentage of observations fall.



For example, the 90th percentile is the value below which 90 percent of the observations may be found. And it is also the value above which 10 percent of the observations may be found.

- Percentiles are commonly used to report scores in tests, like the SAT, GRE and LSAT. for example, the 70th percentile on the 2019 GRE was 156. That means if you scored 156 on the exam, your score was better than 70 percent of test takers.

Percentiles- Interpretations:

Percentile: the value below which a percentage of data falls.

Example: You are the fourth tallest person in a group of 20

80% of people are shorter than you:



That means you are at the **80th percentile**.

If your height is 1.85m then "1.85m" is the 80th percentile height in that group.

A score percentile represents the percentage of scores that are equal or below a certain score within a given sample.

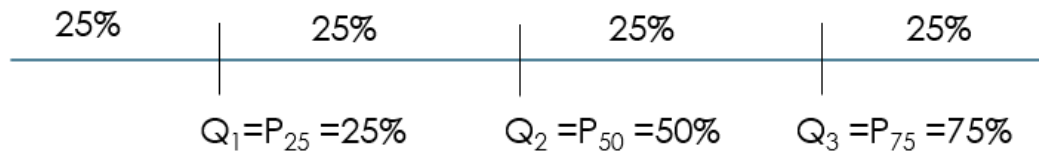
Example: The 75th percentile SAT score for incoming freshmen is 1400.



Relationship:

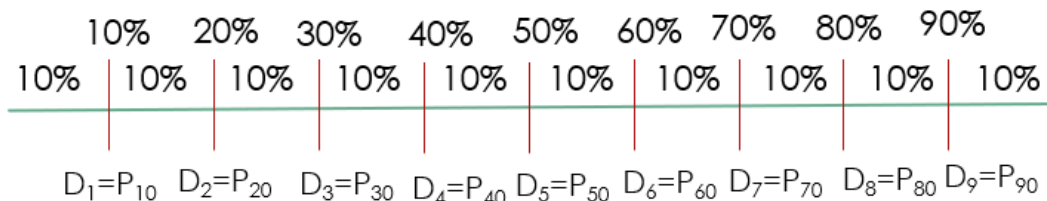
Quartiles:

Quartiles Divides the whole distribution into 4 equal parts



Deciles:

Deciles divide the whole distribution into 10 equal parts



Deciles		Percentiles
D_1	=	P_{10}
D_2	=	P_{20}
D_3	=	P_{30}
D_4	=	P_{40}
D_5	=	P_{50}
D_6	=	P_{60}
D_7	=	P_{70}
D_8	=	P_{80}
D_9	=	P_{90}

Quartiles		Percentiles
Q_1	=	P_{25}
Q_2	=	P_{50}
Q_3	=	P_{75}

	Quartiles	Deciles	Percentiles
Median	$= Q_2$	$= D_5$	$= P_{50}$

Formulas for finding percentiles:

For raw data,

- If $\frac{in}{100}$ is an integer- (for $i = 1, 2, 3, \dots, 99$)

$$P_i = \frac{1}{2} \left[\left(\frac{in}{100} \right)^{th} \text{ value} + \left(\frac{in}{100} + 1 \right)^{th} \text{ value} \right]$$

- If $\frac{in}{100}$ is not an integer-

$$P_i = \text{next integer}^{th} \text{ value of } \frac{in}{100}$$

Formulas for finding quartiles:

For raw data,

- If $\frac{in}{4}$ is an integer- (for $i = 1, 2, 3$)

$$Q_i = \frac{1}{2} \left[\left(\frac{in}{4} \right)^{th} \text{ value} + \left(\frac{in}{4} + 1 \right)^{th} \text{ value} \right]$$

- If $\frac{in}{4}$ is not an integer-

$$Q_i = \text{next integer}^{th} \text{ value of } \frac{in}{4}$$

Formulas for finding deciles:

- If $\frac{in}{10}$ is an integer- (for $i = 1, 2, 3, \dots, 9$)

$$D_i = \frac{1}{2} \left[\left(\frac{in}{10} \right)^{th} \text{ value} + \left(\frac{in}{10} + 1 \right)^{th} \text{ value} \right]$$

- If $\frac{in}{10}$ is not an integer-

$$D_i = \text{next integer}^{th} \text{ value of } \frac{in}{10}$$

Example 1:

Weekly income of 6 respondents (in taka)-

2500, 3900, 3500, 5000, 4000, 3500

Find Q1, Q2, Q3.

So, the quartiles- (n=6)

$$Q_1 = P_{25} = \text{next integer}^{th} \text{value of } \frac{25n}{100} = \text{next integer}^{th} \text{value of } \frac{150}{100} \\ = \text{next integer}^{th} \text{value of } 1.5 = 2^{nd} \text{ value} = 3500$$

$$Q_2 = P_{50} = \text{next integer}^{th} \text{value of } \frac{50n}{100} = \text{next integer}^{th} \text{value of } \frac{300}{100} = 3^{rd} \text{ value}$$

$$Q_2 = P_{50} = \frac{1}{2} \left[\left(\frac{n}{100} \right)^{th} \text{ value} + \left(\frac{n}{100} + 1 \right)^{th} \text{ value} \right] = \frac{1}{2} [3^{rd} \text{ value} + 4^{th} \text{ value}] \\ = \frac{3500 + 3900}{2} = 3700$$

Or

$$Q_2 = P_{50} = \text{Median} = \frac{1}{2} \left[\left(\frac{n}{2} \right)^{th} \text{ value} + \left(\frac{n}{2} + 1 \right)^{th} \text{ value} \right] = 3700$$

$$Q_3 = P_{75} = \text{next integer}^{th} \text{value of } \frac{75n}{100} = \text{next integer}^{th} \text{value of } \frac{450}{100} \\ = \text{next integer}^{th} \text{value of } 4.5 = 5^{th} \text{ value} = 4000$$

Example 2:

Suppose we had a class of 15 students who took a math test there were 50 questions on the math test and the next day everyone got their scores back. The scores are given below

34, 20, 35, 39, 38, 25, 43, 36, 31, 36, 35, 38, 15, 34, 36

Find

1. 50th Percentile or 2nd Quartiles or 5th Deciles
2. 1st Quartiles or 25th Percentile
3. 3rd Quartiles or 75th Percentile

We have to arrange the data in ascending order-

15, 20, 25, 31, 34, 34, 35, 35, 36, 36, 36, 38, 38, 39, 43

8th position

4th position

12th position

So, the percentiles-

$$Q_2 = D_5 = P_{50} = \text{next integer}^{th} \text{ value of } \frac{50n}{100} = \text{next integer}^{th} \text{ value of } \frac{750}{100} \\ = \text{next integer}^{th} \text{ value of } 7.5 = 8th \text{ value} = 35$$

Comments:

50% of the data points are below or equal to 35 & 50% of the data points are over 35

$$Q_1 = P_{25} = \text{next integer}^{th} \text{ value of } \frac{25n}{100} = \text{next integer}^{th} \text{ value of } \frac{375}{100} \\ = \text{next integer}^{th} \text{ value of } 3.75 = 4th \text{ value} = 31$$

Comments:



25% of the data points are below or equal to 31 & 75% of the data points are over 31

$$Q_3 = P_{75} = \text{next integer}^{th} \text{value of } \frac{75n}{100} = \text{next integer}^{th} \text{value of } \frac{1125}{100} \\ = \text{next integer}^{th} \text{value of } 11.25 = 12^{th} \text{ value} = 38$$

Comments:

75% of the data points are below or equal to 38 & 25% of the data points are over 38

or If someone scored 75th percentile you can say that s/he beat 75% of the class.

Example 3:

► Now look at the following arranged data set (n = 20)

3	3	3	4	5	5	6	7	8	9	11	12	13	14	17	20	21	21	23	25
				↑	↑				↑	↑				↑	↑				
				5 th	6 th				10 th	11 th				15 th	16 th				
				Positions					Positions					Positions					

$$Q_1 = P_{25} = \frac{(25 \times 20)}{100} = \frac{500}{100} = 5^{th} \text{ position value} = \frac{1}{2} [5^{th} \text{ value} + 6^{th} \text{ value}] = \frac{5+5}{2} = \underline{5}$$

$$Q_2 = P_{50} = \frac{(50 \times 20)}{100} = \frac{1000}{100} = 10^{th} \text{ position value} = \frac{1}{2} [10^{th} \text{ value} + 11^{th} \text{ value}] = \frac{9+11}{2} = \underline{10}$$

$$Q_3 = P_{75} = \frac{(75 \times 20)}{100} = \frac{1500}{100} = 15^{th} \text{ position value} = \frac{1}{2} [15^{th} \text{ value} + 16^{th} \text{ value}] = \frac{17+20}{2} = \underline{18.5}$$